

A photograph of several industrial smokestacks emitting dark smoke, set against a hazy sunset sky. The sun is a bright orange orb in the center, with its reflection shimmering on the calm water in the foreground. The smokestacks are silhouetted against the bright sky.

# GEN203 Ecological System and Environment

Week 6: Environmental Pollution  
[Ekhtekharul.islam@ewubd.edu](mailto:Ekhtekharul.islam@ewubd.edu)

# Topics and Learning Outcomes

## Topics

- Environmental Pollution – Basic Terms
- Types of Environmental Pollution
- Air Pollution – Types
- Air Quality Index
- Major Air Pollutants
- Consequences of Air Pollution

## Learning Outcomes

- After this lesson, students will be able to:
  - Define and correctly use each of the key concepts
  - Give examples of different types of sources of air, and water pollution.
  - Explain the relationship between current practices that result in environmental pollution.

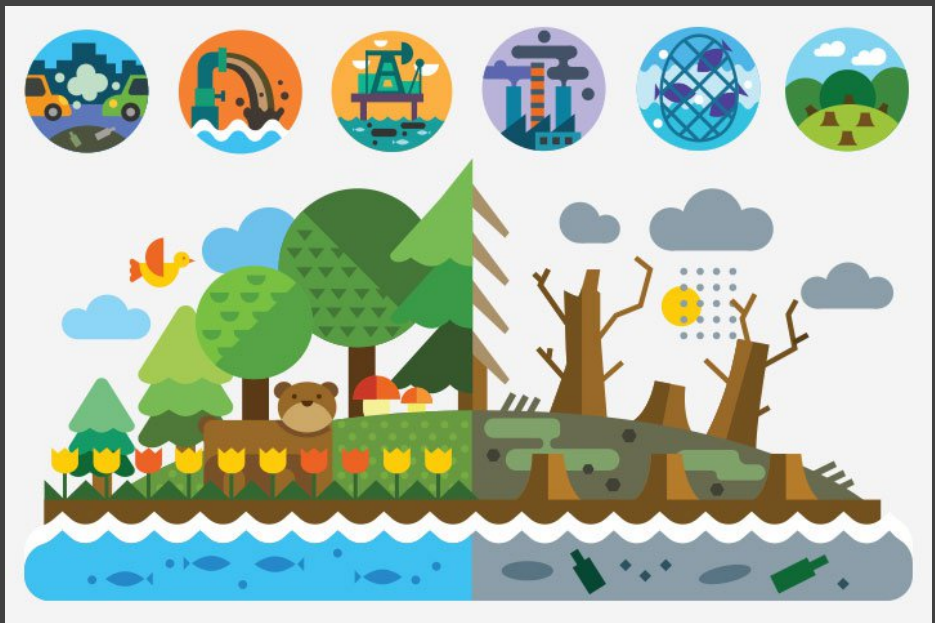
# Environmental Pollution - Terminologies

**Environmental pollution:** Environmental pollution is an effect caused by undesirable changes in the natural environment that have a harmful effect on plants, animals and humans.

**Pollutant:** Any substance causing **nuisance or harmful effects** or uneasiness to the organisms

Pollutants can be **naturally occurring** substances or energies, but they are considered contaminants when in excess of natural levels.

Pollutants can **decompose and are rapidly destroyed** by natural processes similar to the decomposition of vegetables. We also have pollutants that take **decades to decompose and are difficult to remove** once released. DDT, plastics, heavy metals and nuclear waste are just a few example



# Types of Pollution

WATER POLLUTION

AIR POLLUTION

LAND POLLUTION

NOISE POLLUTION

THERMAL POLLUTION

LIGHT POLLUTION

VISUAL POLLUTION



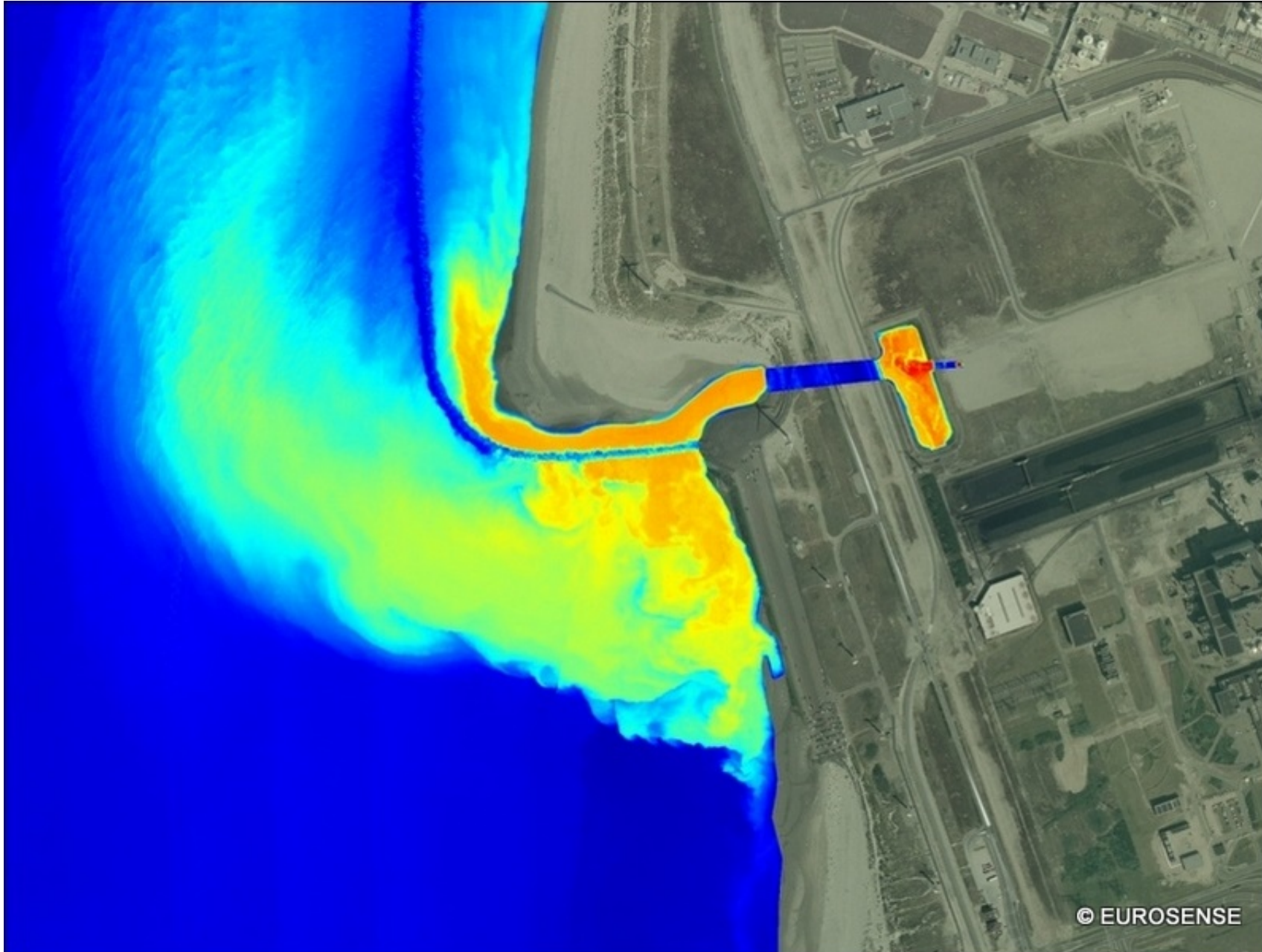
#ToxicCities  
HEAR ME?

She  
the  
People TV



# Noise Pollution

**Noise pollution**, also known as environmental **noise** or **sound pollution**, is the propagation of **noise** with harmful impact on the activity of human or animal life



**Thermal pollution** is the degradation of water quality by any process that changes ambient water temperature.

# UNDERSTANDING LIGHT POLLUTION



**Light pollution**, also known as photo **pollution**, is the presence of anthropogenic and artificial **light** in the night environment.



# Visual Pollution

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**Visual pollution** is an aesthetic issue and refers to the impacts of **pollution** that impair one's ability to enjoy a vista or view.

# Emerging Environmental Problems



<https://www.youtube.com/watch?v=tLfNUd0-8ts>

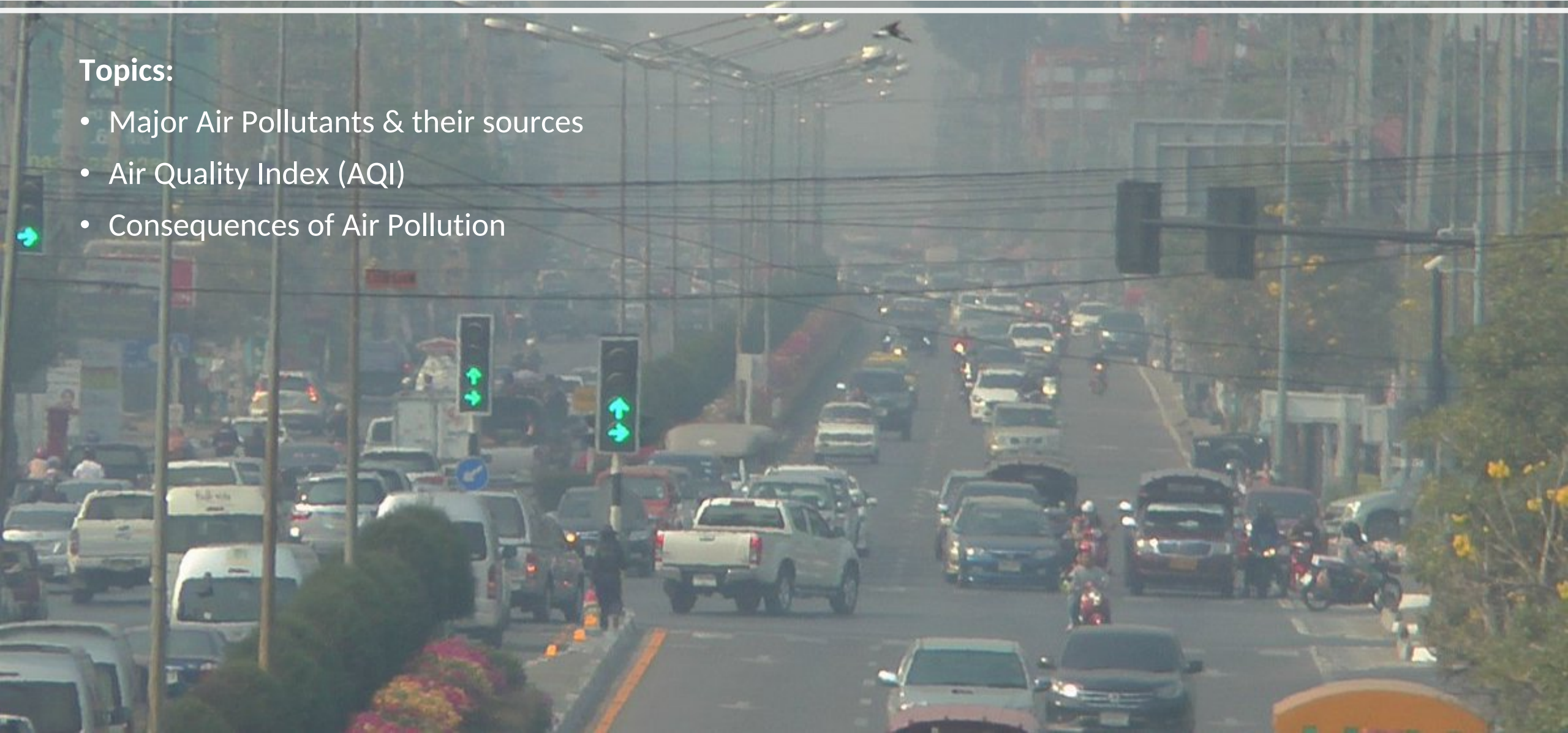


<https://www.youtube.com/watch?v=PMfdCeVyYsA&t=2s>

# AIR POLLUTION

## Topics:

- Major Air Pollutants & their sources
- Air Quality Index (AQI)
- Consequences of Air Pollution



# Bangladesh loses \$14bn a year to air pollution

An estimated 96,000 Bangladeshi children died before their fifth birthday in 2018 from PM2.5 pollution



[https://tbsnews.net/environment/bangladesh-loses-14bn-year-air-pollution-44359?fbclid=IwAR3gggFjLaKvPH-qvVyUor2KhF4cOiV\\_FMOhN-mWi3EZp\\_zSHrq351NfHaE](https://tbsnews.net/environment/bangladesh-loses-14bn-year-air-pollution-44359?fbclid=IwAR3gggFjLaKvPH-qvVyUor2KhF4cOiV_FMOhN-mWi3EZp_zSHrq351NfHaE)

Photo: Saikat Bhadra/TBS

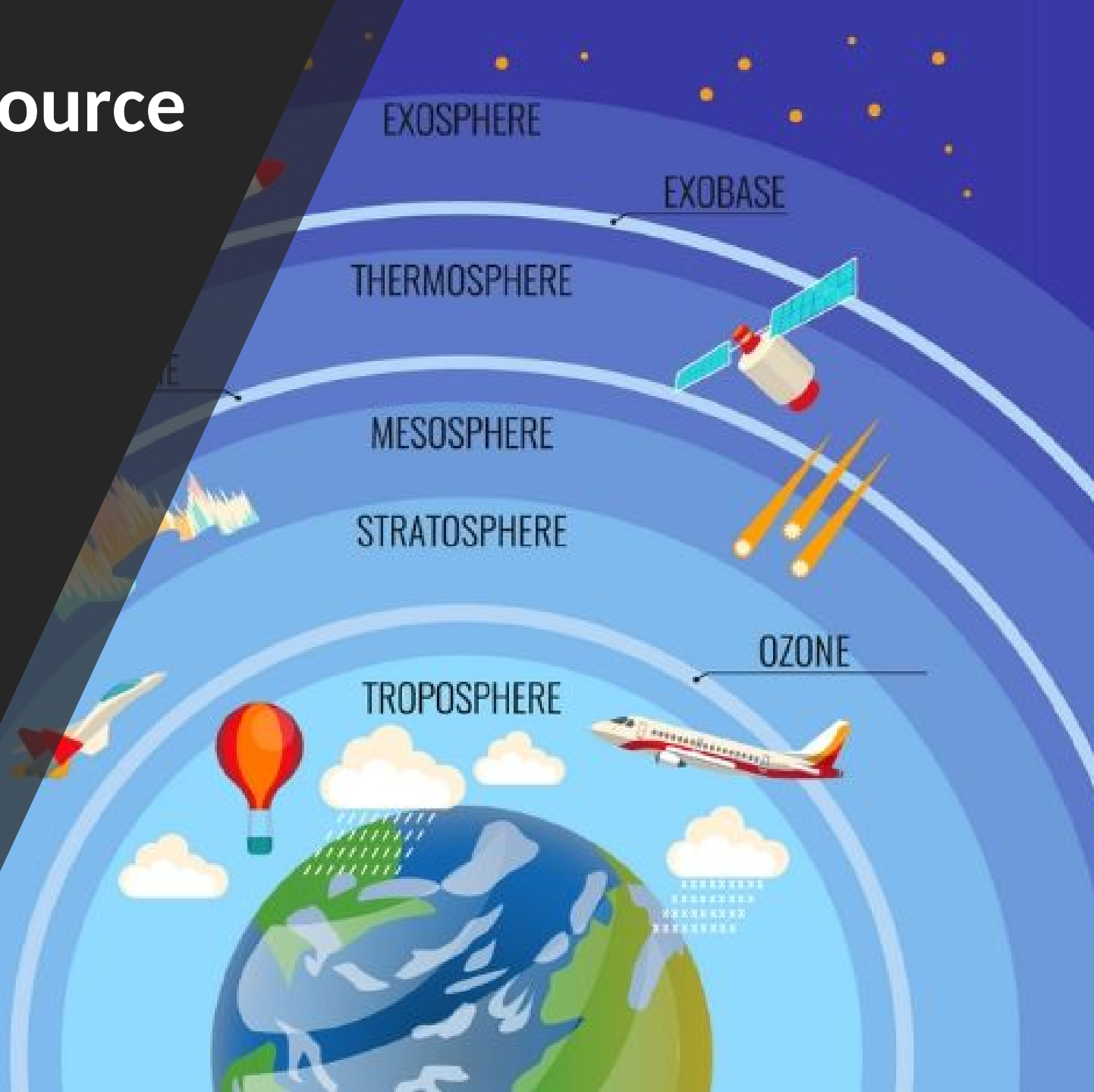


From Facebook

## Air Pollution Sources

# Atmosphere as a Resource

- The Earth is surrounded by a ***blanket of air***, which we call the atmosphere.
- It reaches over 560 kilometers (348 miles) from the surface of the Earth, so we are only able to see what occurs fairly close to the ground.
- This “blanket”
  - moderates temperature
  - protects us from harmful radiation and
  - supports life.



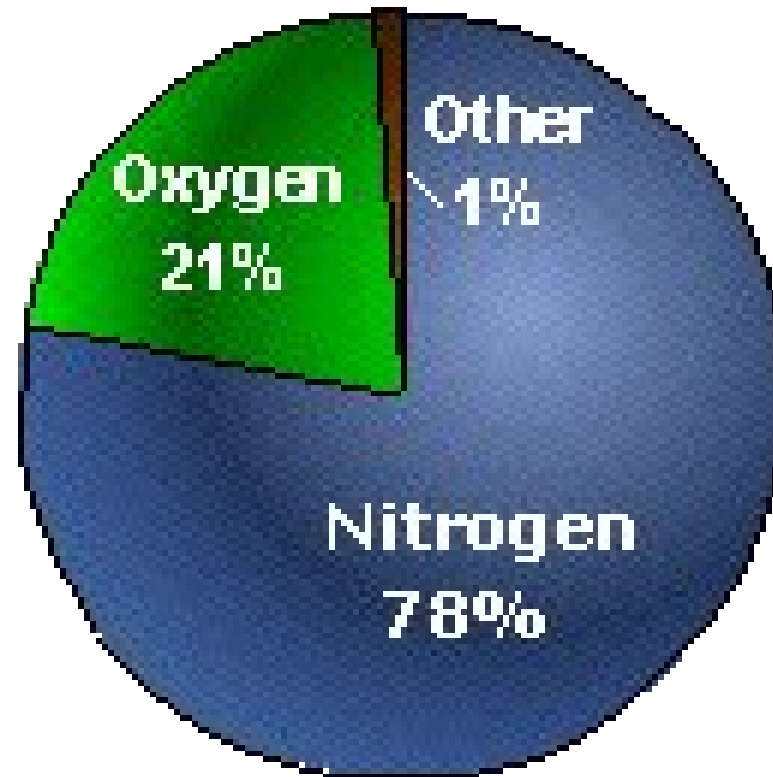
# Composition of the Atmosphere

The present composition of the atmosphere is:

- 21% ☾ OXYGEN ( $O_2$ )
- 78% ☾ NITROGEN ( $N_2$ )
- 0.04% ☾ CARBON DIOXIDE ( $CO_2$ )
- ~0.9% ☾ ARGON (Ar)

Besides water vapor, several other gases are also present in much smaller amounts:

- Carbon monoxide (CO)
- Neon (Ne)
- Methane ( $CH_4$ )
- Krypton (Kr)



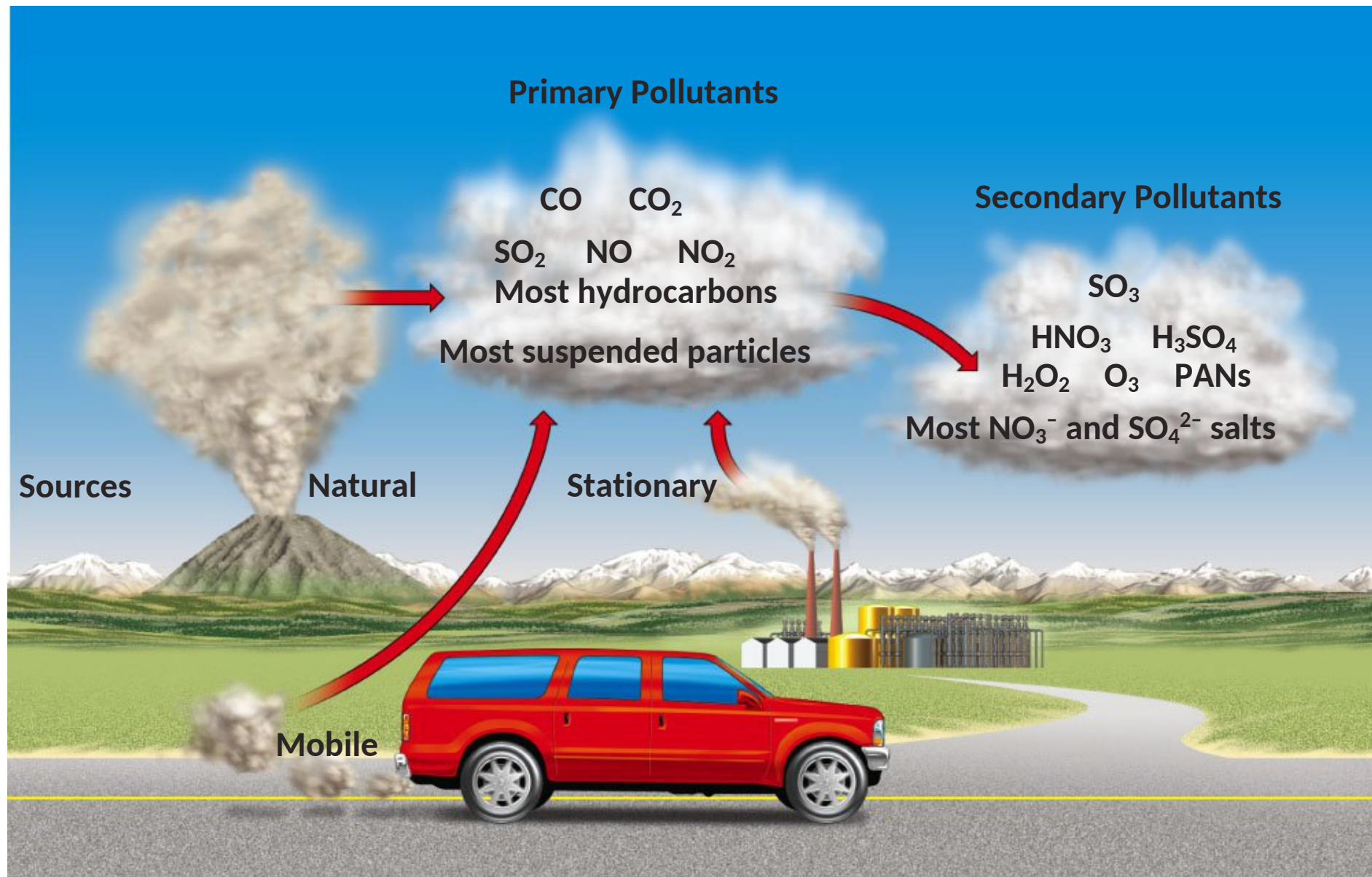
What if this composition changes?



# Types and Sources of Air Pollution

- **Air Pollution**
  - **Chemicals** added to the atmosphere by **natural events** or **human activities** in high enough concentrations to be harmful
- **Two categories of pollutant**
  - A. **Primary Air Pollutant**
    - Harmful substance that is emitted **directly** into the atmosphere
  - B. **Secondary Air Pollutant**
    - Harmful substance formed in the atmosphere when a **primary air pollutant** reacts with **substances** normally found in the atmosphere or with other air pollutants



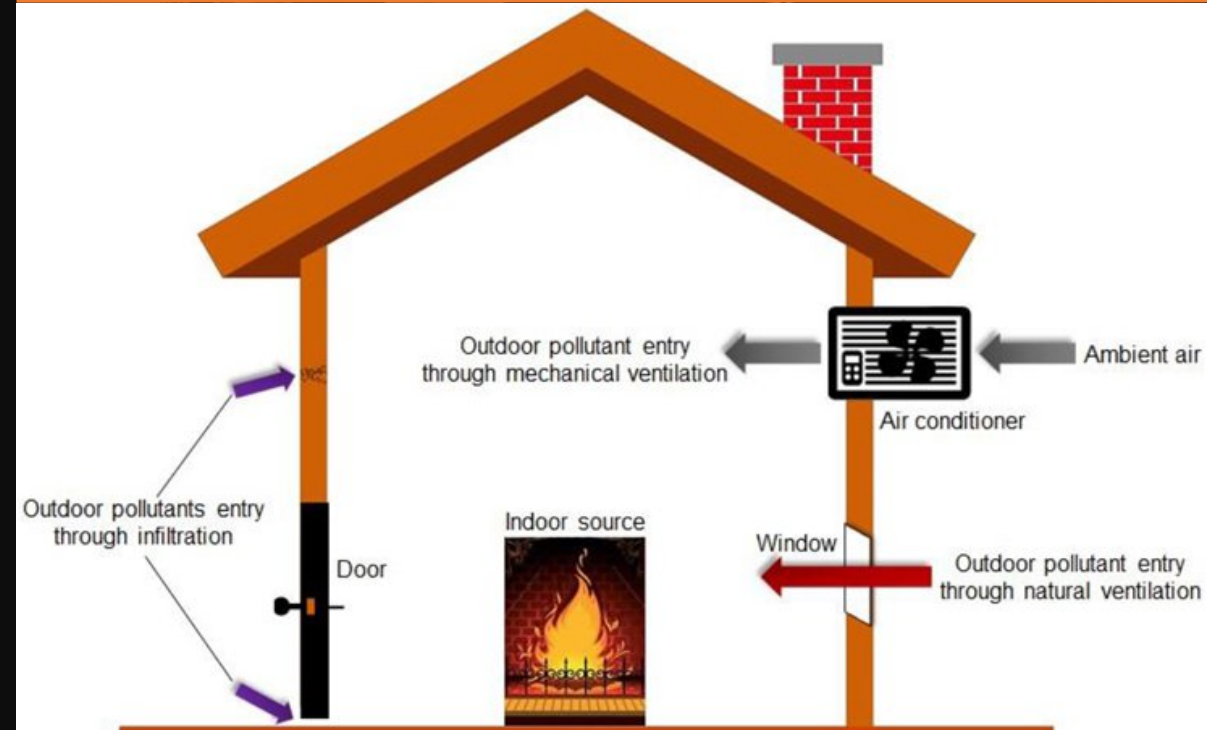


© 2007 Thomson Higher Education

Fig. 19-3, p. 442

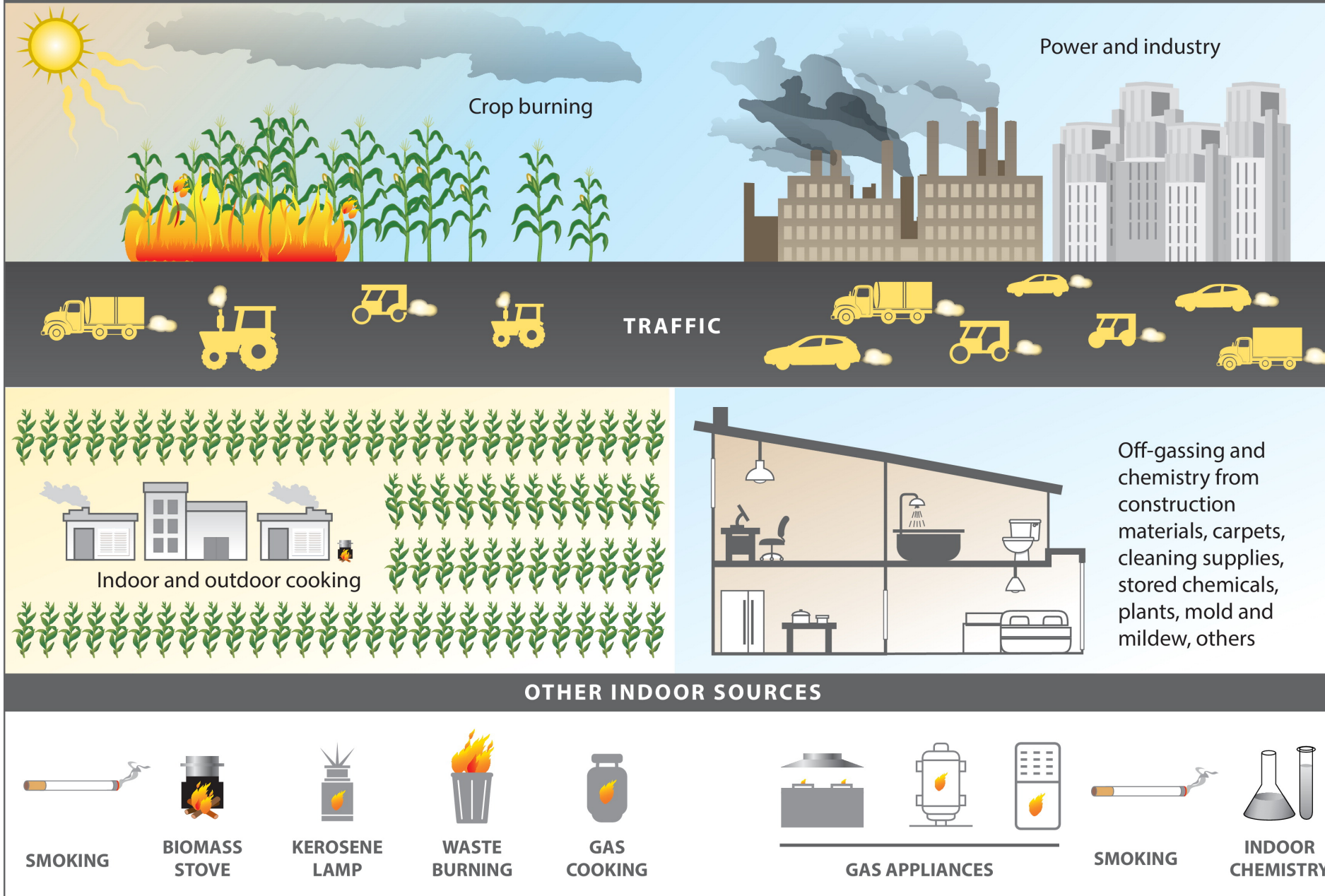
# Types of Air Pollution

- **Indoor air pollution**: Tobacco smoke, cooking and heating appliances, and vapors from building materials, paints, furniture, etc. cause pollution inside buildings
- **Outdoor air pollution**: chemical reactions between pollutants derived from different sources, primarily automobile exhaust and industrial emissions



## RURAL SETTINGS

## URBAN SETTINGS





## Dhaka City's Pollution Sources

New York City  
Monday



New York City  
Wednesday



Bladerunner  
2049





This Photo by Unknown Author is licensed under [CC BY-SA](#)



**China** is buying fresh air from Rocky Mountain  
Area of **Canada**

<https://www.youtube.com/watch?v=LblevNlt-JE>

# Anthropogenic Sources of Air Pollutants



Emissions from  
factories, cars, planes,  
or aerosol cans



Second-  
hand cigarette smoke

Pollution enters the Earth's atmosphere in **many different ways**.

Air pollution is **most common in large cities** where emissions from many different sources are concentrated.

Sometimes, **mountains or tall buildings** prevent air pollution from spreading out

Large cities in **poor and developing nations** tend to have more air pollution than cities in developed nations.

# How are these pollutants spread?

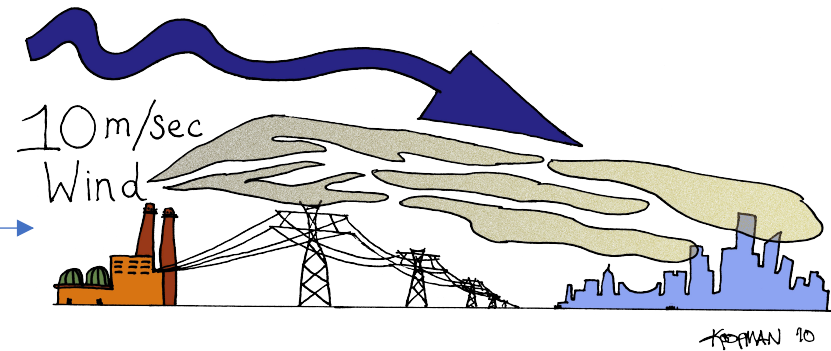
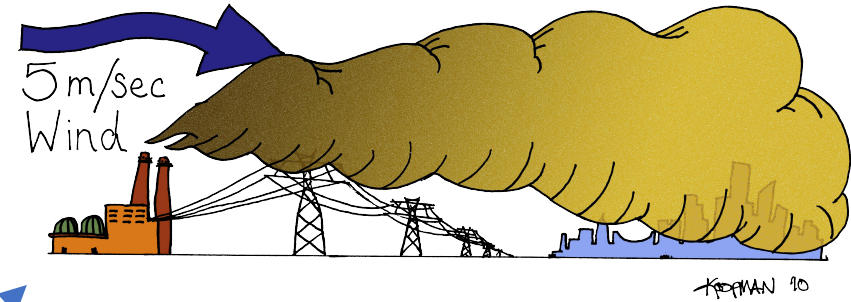
## Wind

**“Dilution is the solution to pollution”**

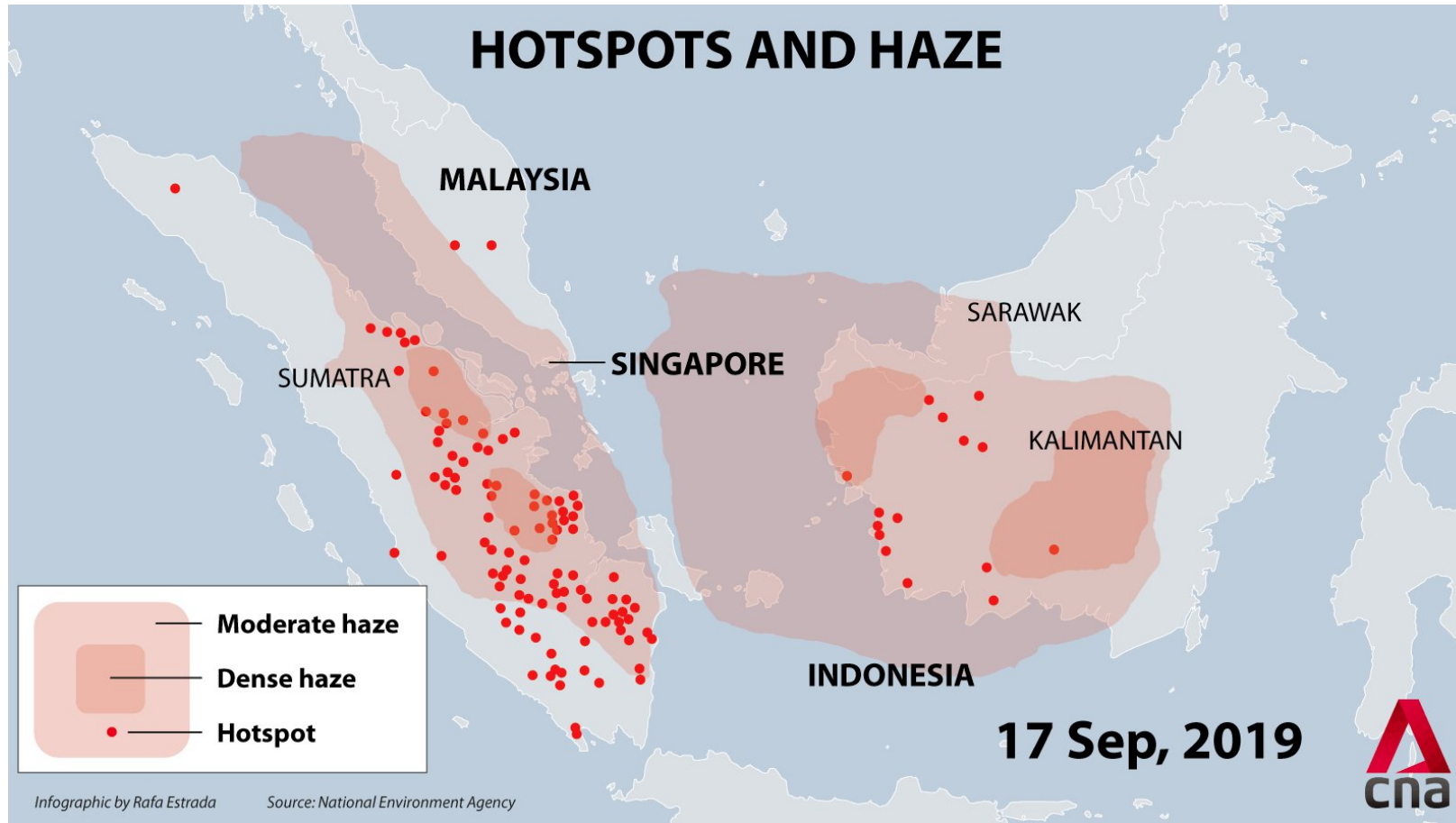
Wind causes bodies of polluted air to spread out across the Earth.

**When winds are low**, these bodies sit and stagnate over a smaller area leading to localized pockets of more concentrated pollution.

**When winds are high**, these bodies are spread out over larger areas, leading to a lower concentration of pollutants.



# Air Pollution in Indonesia



# MAJOR AIR POLLUTANTS

**Table 20.1 Major Air Pollutants**

| <i>Pollutant</i>                 | <i>Composition</i>             | <i>Primary or Secondary</i> | <i>Characteristics</i>         |
|----------------------------------|--------------------------------|-----------------------------|--------------------------------|
| <b><i>Particulate matter</i></b> |                                |                             |                                |
| Dust                             | Variable                       | Primary                     | Solid particles                |
| Lead                             | Pb                             | Primary                     | Solid particles                |
| Sulfuric acid                    | H <sub>2</sub> SO <sub>4</sub> | Secondary                   | Liquid droplets                |
| <b><i>Nitrogen oxides</i></b>    |                                |                             |                                |
| Nitrogen dioxide                 | NO <sub>2</sub>                | Primary                     | Reddish-brown gas              |
| <b><i>Sulfur oxides</i></b>      |                                |                             |                                |
| Sulfur dioxide                   | SO <sub>2</sub>                | Primary                     | Colorless gas with strong odor |
| <b><i>Carbon oxides</i></b>      |                                |                             |                                |
| Carbon monoxide                  | CO                             | Primary                     | Colorless, odorless gas        |
| Carbon dioxide*                  | CO <sub>2</sub>                | Primary                     | Colorless, odorless gas        |
| <b><i>Hydrocarbons</i></b>       |                                |                             |                                |
| Methane                          | CH <sub>4</sub>                | Primary                     | Colorless, odorless gas        |
| Benzene                          | C <sub>6</sub> H <sub>6</sub>  | Primary                     | Liquid with sweet smell        |
| <b><i>Ozone</i></b>              |                                |                             |                                |
|                                  | O <sub>3</sub>                 | Secondary                   | Pale blue gas with acrid odor  |
| <b><i>Air toxics</i></b>         |                                |                             |                                |
| Chlorine                         | Cl <sub>2</sub>                | Primary                     | Yellow-green gas               |

\* Discussed in Chapter 21.

Source: Environmental Protection Agency.

# Particulate Matter

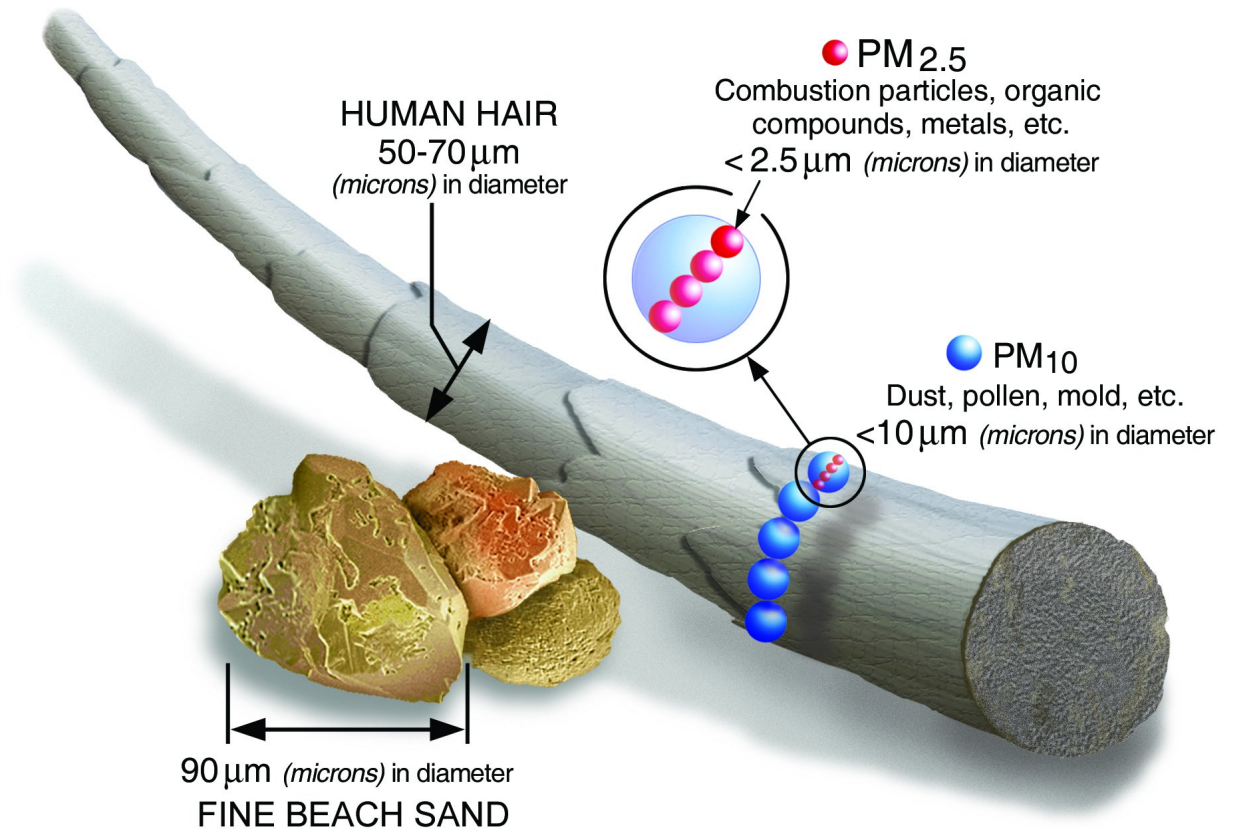
- Sea salt, soil dust, volcanic particles, smoke from forest fires account for particulate emissions each year.
- Small particles are removed from the atmosphere by accretion to water droplets, which grow in size until they are large enough to precipitate.
- Larger particles are removed by direct washout by falling raindrops.



# Particulate Matter

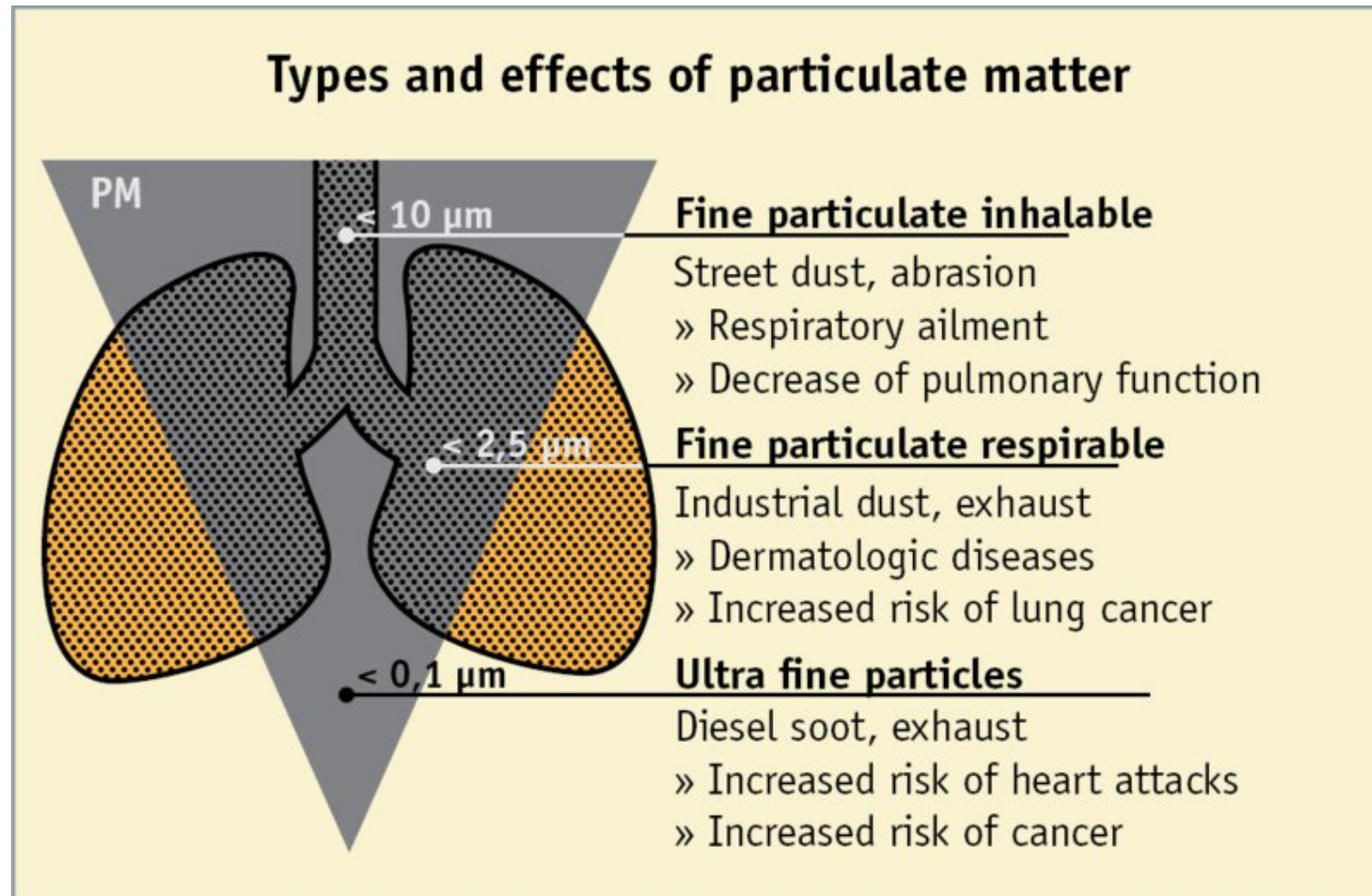
Particulate Matters are divided into **two categories**:

- "**Inhalable coarse particles**," such as those found near roadways and dusty industries, are larger than 2.5 micrometers and smaller than 10 micrometers in diameter.
- "**Fine particles**," such as those found in smoke and haze, are 2.5 micrometers in diameter and smaller. These particles can be directly emitted from sources such as forest fires, or they can form when gases emitted from power plants, industries and automobiles react in the air.



# Particulate Matter

Particulate matter contains microscopic solids or liquid droplets that are so small that they can be inhaled and cause serious health problems.



# Major Air Pollutants (1)

- *Carbon oxides:*

- Carbon monoxide (CO) is a **highly toxic gas** that forms during the incomplete combustion of carbon-containing materials.
- 93% of carbon dioxide (CO<sub>2</sub>) in the troposphere occurs as a result of the carbon cycle.
- 7% of CO<sub>2</sub> in the troposphere occurs as a result of human activities (mostly burning fossil fuels).
  - It is not regulated as a pollutant under the U.S. Clean Air Act.

# Major Air Pollutants (2)

- *Nitrogen oxides* and *nitric acid*:

- Nitrogen oxide (NO) forms when nitrogen and oxygen gas in air react at the high-combustion temperatures in automobile engines and coal-burning plants.
- NO can also form from lightening and certain soil bacteria.
  - NO reacts with air to form NO<sub>2</sub>.
  - NO<sub>2</sub> reacts with water vapor in the air to form nitric acid (HNO<sub>3</sub>) and nitrate salts (NO<sub>3</sub><sup>-</sup>) which are components of acid deposition.

# Major Air Pollutants (3)

- *Sulfur dioxide (SO<sub>2</sub>)* and *sulfuric acid*:
  - About **one-third of SO<sub>2</sub>** in the troposphere occurs **naturally**.
  - **Two-thirds come from human sources**, mostly combustion ( $S + O_2 \rightarrow SO_2$ ) of sulfur-containing coal and from oil refining and smelting of sulfide ores.
  - SO<sub>2</sub> in the atmosphere can be converted to **sulfuric acid (H<sub>2</sub>SO<sub>4</sub>)** and **sulfate salts (SO<sub>4</sub><sup>2-</sup>)** that return to earth as a component of acid deposition.

# Major Air Pollutants (4)

- *Suspended particulate matter (SPM):*

- Consists of a variety of **solid particles** and **liquid droplets** small and light enough to remain suspended in the air.
- The most harmful forms of SPM are **fine particles** (PM-10, with an average diameter < 10 micrometers) and **ultrafine particles** (PM-2.5).
- According to the EPA, SPM is responsible for about **60,000 premature deaths** a year in the U.S.

# How do we measure the **QUALITY** of AIR?



## AQI: Air Quality Index

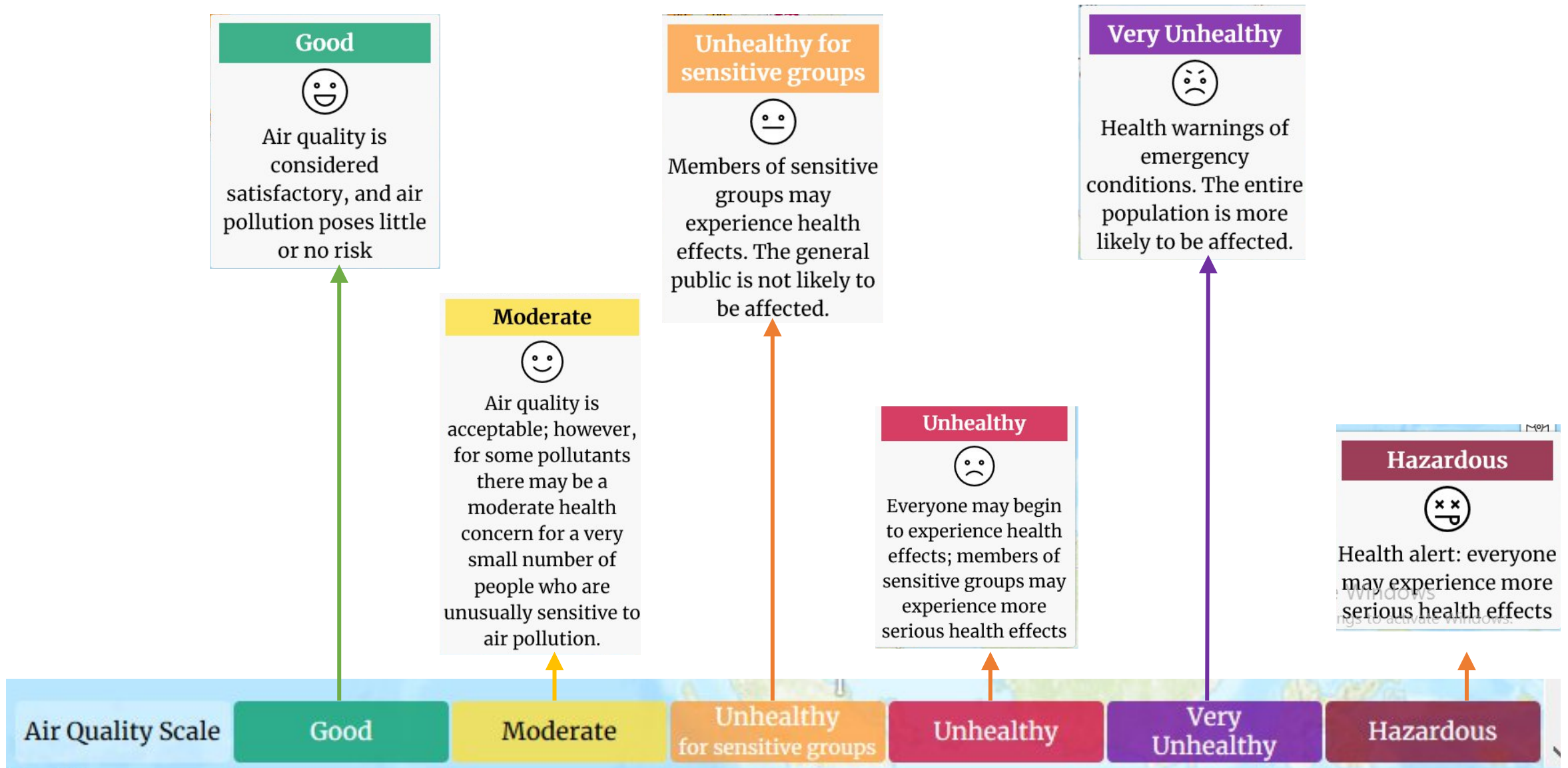
- Indicates whether pollutant levels in air may cause health concerns.
- is used by government agencies to communicate to the public how polluted the air currently is or how polluted it is forecast to become.
- Ranges from 0 (least concern) to 500 (greatest concern)

<https://waqi.info/>

<https://images.app.goo.gl/95F6CfabQivNgQN7>

*Air quality is determined by its level of cleanness*

# Air Quality Index - Scale

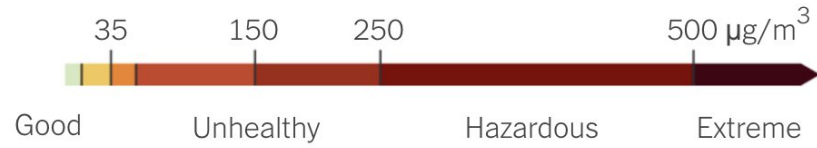


# AQI: Air Quality Index

| AQI color code     | Who is affected?                                                                                      | What is the significance?                                                        | What action should people take?                                                                            |
|--------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Green (0-50)       | –                                                                                                     | Air quality is good                                                              | Enjoy activities                                                                                           |
| Yellow (51-100)    | People who are unusually sensitive to air pollution                                                   | Air quality is a concern for people who are unusually sensitive to air pollution | People unusually sensitive to air pollution: Plan strenuous activities when air quality is better          |
| Orange (101 – 150) | People with heart or lung disease (including asthma), older adults, and children                      | Air quality is unhealthy for people in sensitive groups                          | Sensitive groups: Cut back or reschedule strenuous activities                                              |
| Red (151-200)      | Everyone, especially people with heart or lung disease (including asthma), older adults, and children | Air quality is unhealthy for everyone                                            | Everyone: Cut back or reschedule strenuous activities<br>Sensitive groups: Avoid strenuous activities      |
| Purple (201-300)   | Everyone, especially people with heart or lung disease (including asthma), older adults, and children | Air quality is very unhealthy for everyone                                       | Everyone: Significantly cut back on physical activities<br>Sensitive groups: Avoid all physical activities |

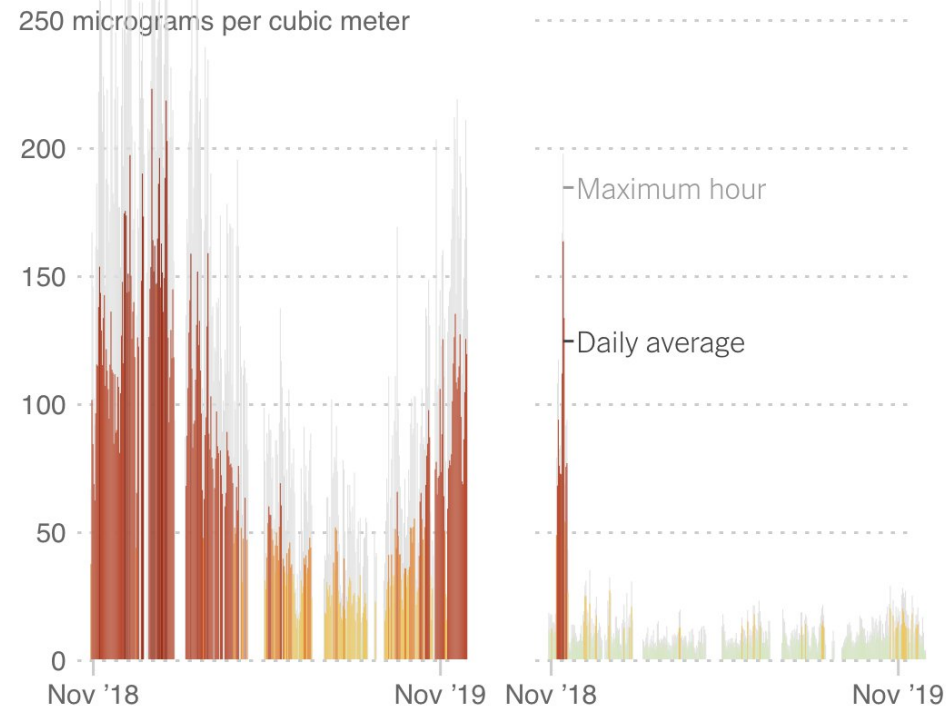
Source: <https://www.epa.gov/pmcourse/patient-exposure-and-air-quality-index>

### DAILY AIR QUALITY BASED ON PM2.5



**Dhaka** ▼

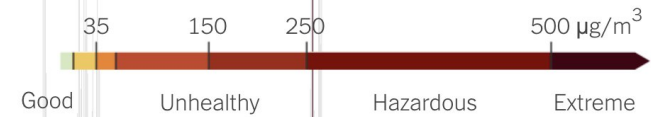
**San Francisco**



[most](#) from unhealthy air.

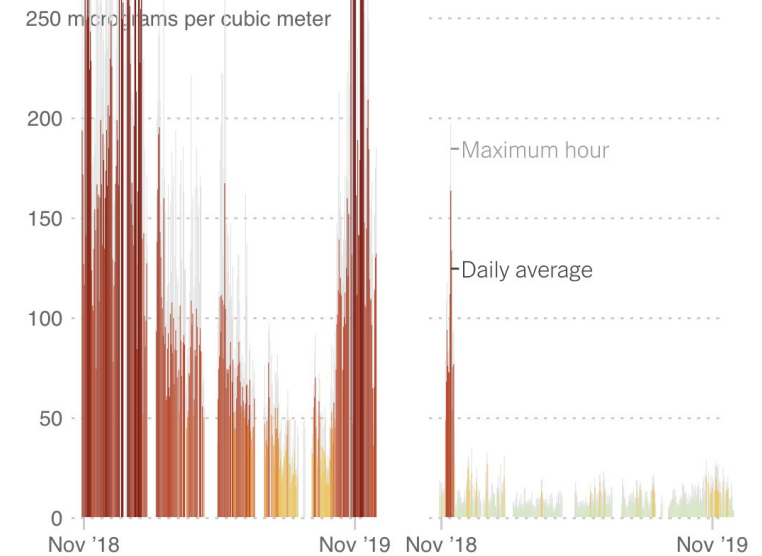
### Wildfires Increase Pollution in California and the Western U.S.

#### DAILY AIR QUALITY BASED ON PM2.5



**Delhi** ▼

**San Francisco**



Data reflects regional estimates by [Berkeley Earth](#) based on



Tweet



**THE SKIN DOCTOR** ✓

@theskindoctor13



Superman after flying through Delhi air for 10 min.



**Sagar**

@sagarcasm



Smoking in Delhi is healthier than doing yoga in Delhi

4 Nov 2022 • 07:17

[www.pikaso.me](http://www.pikaso.me)

# THE COST OF TOXIC AIR



**\$38.3** million lost  
per day due to  
air pollution in  
Bangladesh



China bears the highest  
cost from fossil  
fuel air pollution



**4.5** million die  
prematurely per year  
worldwide due to air pollution



**4** million new cases of  
asthma in children  
each year  
linked to  
**NO2**  
pollution



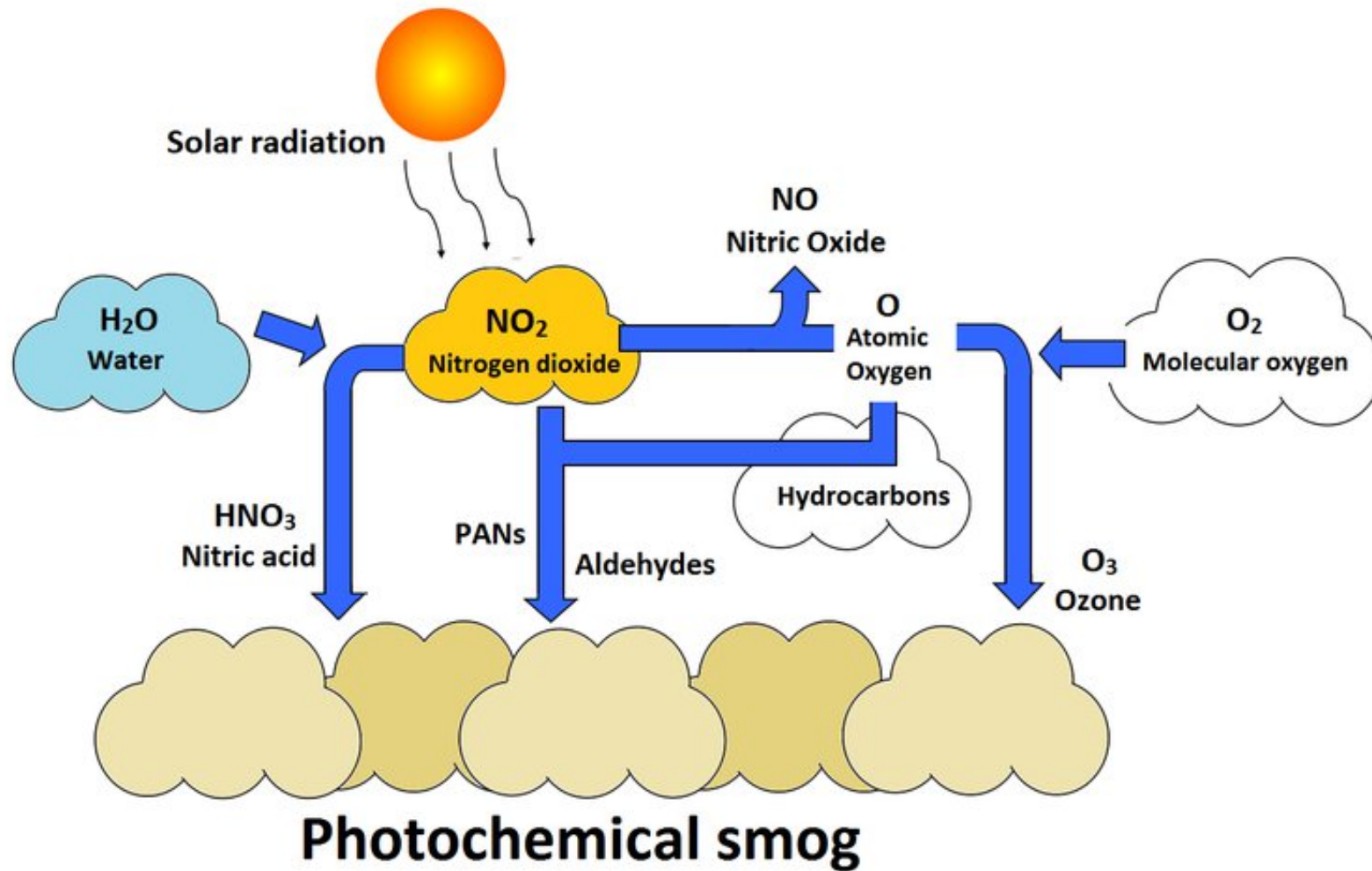
**\$8** billion lost  
worldwide due to  
air pollution from  
fossil fuels every day



|  | Total Cost<br>(Million USD) | Premature<br>Deaths<br>(2018) |
|-------------------------------------------------------------------------------------|-----------------------------|-------------------------------|
| India                                                                               | 150,000                     | 1,000,000                     |
| <b>Bangladesh</b>                                                                   | <b>14,000</b>               | <b>96,000</b>                 |
| Pakistan                                                                            | 6,100                       | 50,000                        |
| Nepal                                                                               | 940                         | 12,000                        |
| Sri Lanka                                                                           | 760                         | 3,300                         |
| Afghanistan                                                                         | 270                         | 3,900                         |
| Bhutan                                                                              | 54                          | 240                           |
| Maldives                                                                            | 26                          | 40                            |

SOURCE: SOURCE: GREENPEACE

# AIR POLLUTANTS cause SMOG

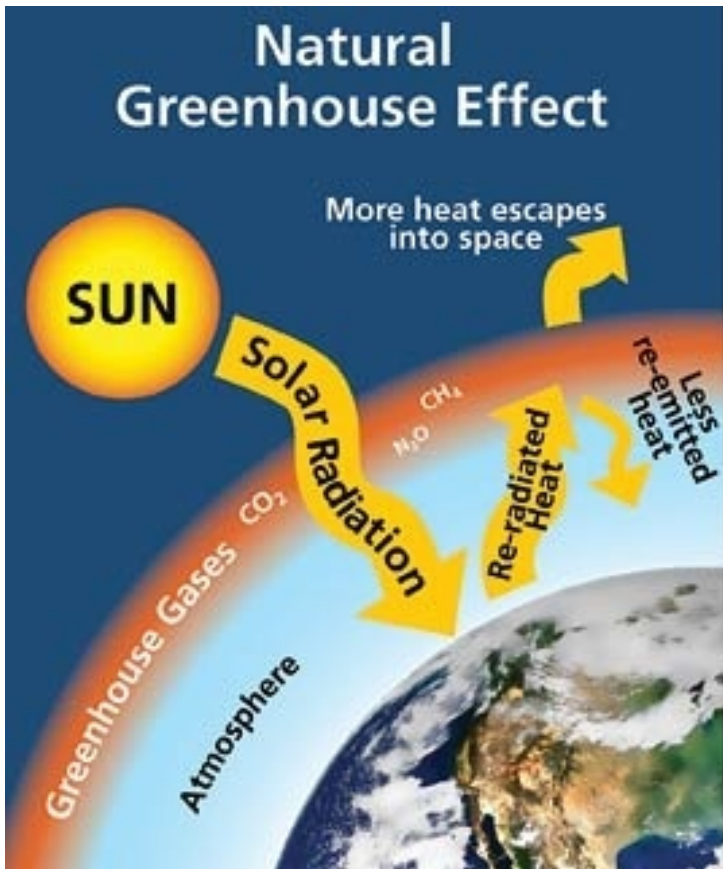


Photochemical smog is a mixture of pollutants that are formed when nitrogen oxides and volatile organic compounds (VOCs) react to sunlight, creating a reddish-brown haze above cities.

It tends to occur more often in summer, because that is when we have the most sunlight.

All modern cities have some photochemical smog, but it is much more common in cities with sunny, warm, and dry climates and lots of motor vehicles

It can cause respiratory diseases such as rhinitis, bronchitis, asthma, pneumonia and other pathologies.



- The Manmade Greenhouse Effect

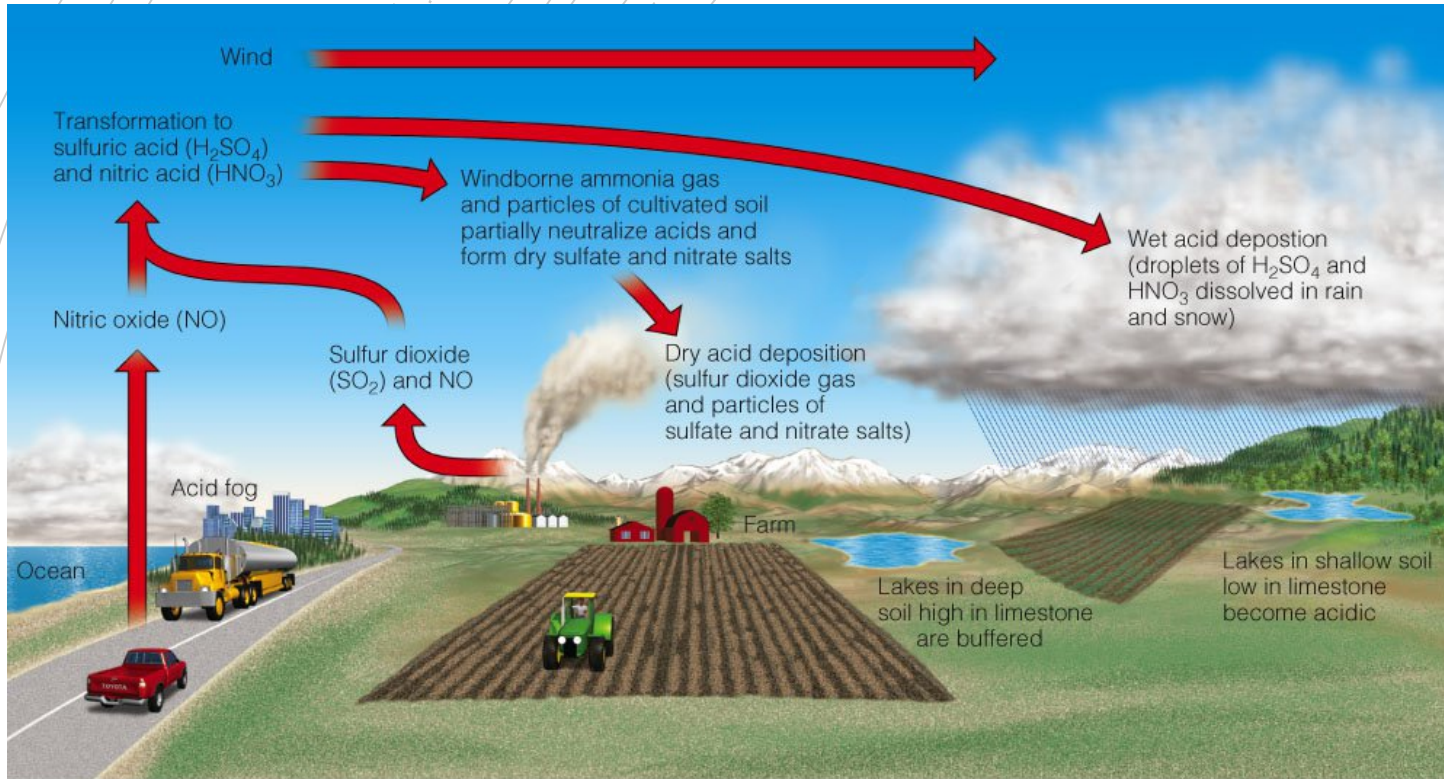
[https://www.youtube.com/watch?v=BPJJM\\_hCFj0](https://www.youtube.com/watch?v=BPJJM_hCFj0)

The **greenhouse effect** is a natural process that warms the Earth's surface. When the Sun's energy reaches the Earth's atmosphere, some of it is reflected back to space and the rest is absorbed and re-radiated by **greenhouse gases**.

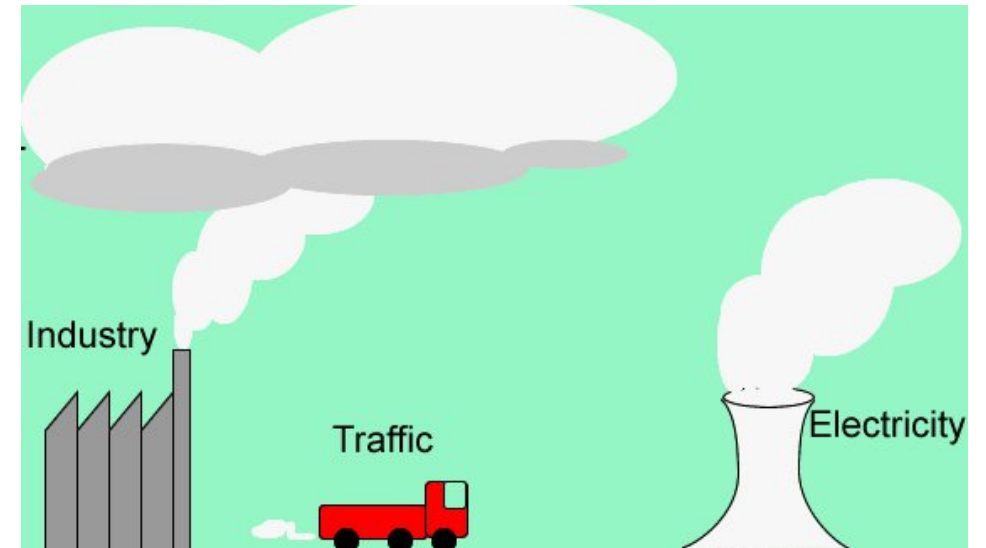
The other **PROBLEM**

- **Acid Rain**

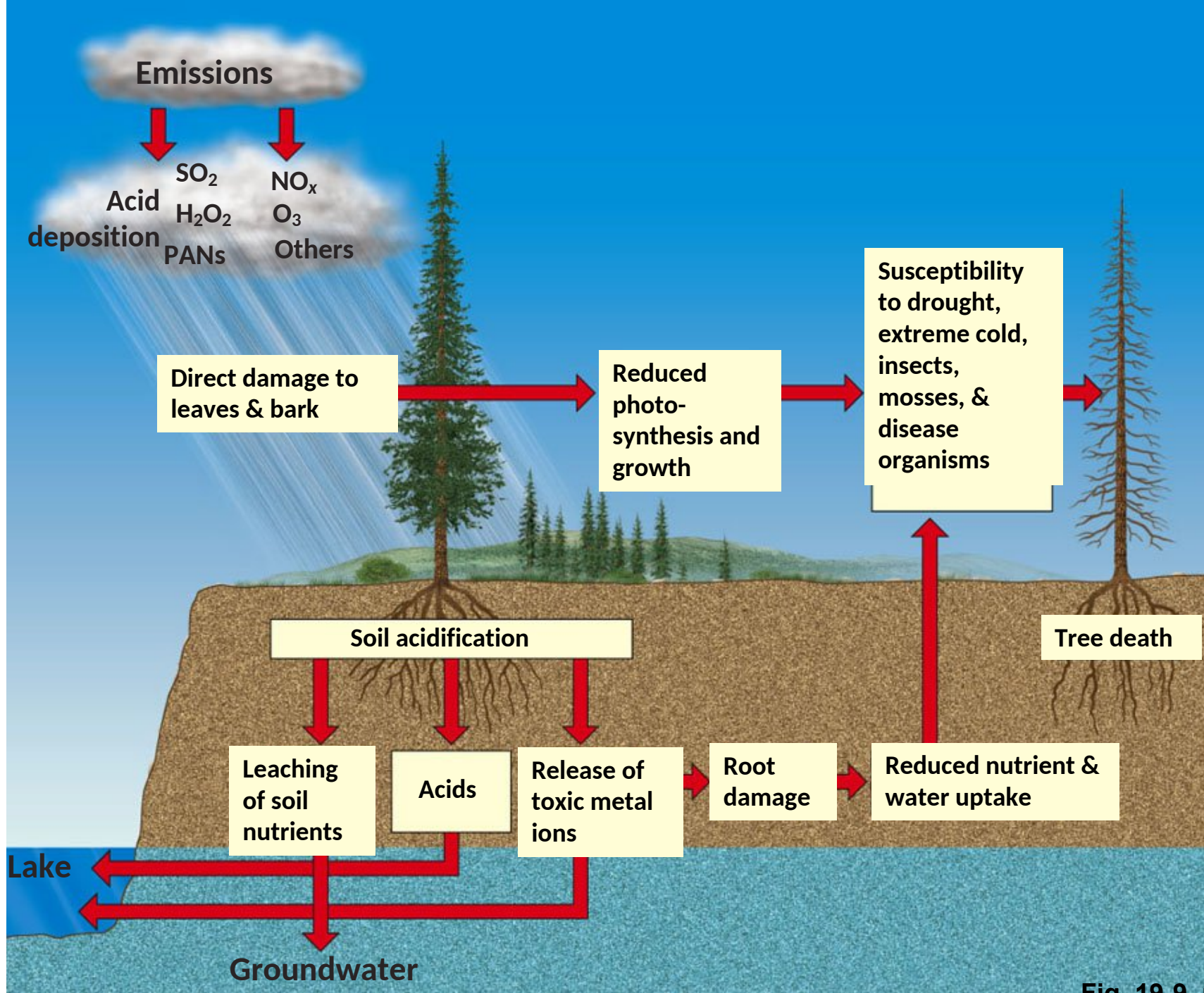
<https://www.youtube.com/watch?v=1PDjVDlrFec>



© 2007 Thomson Higher Education



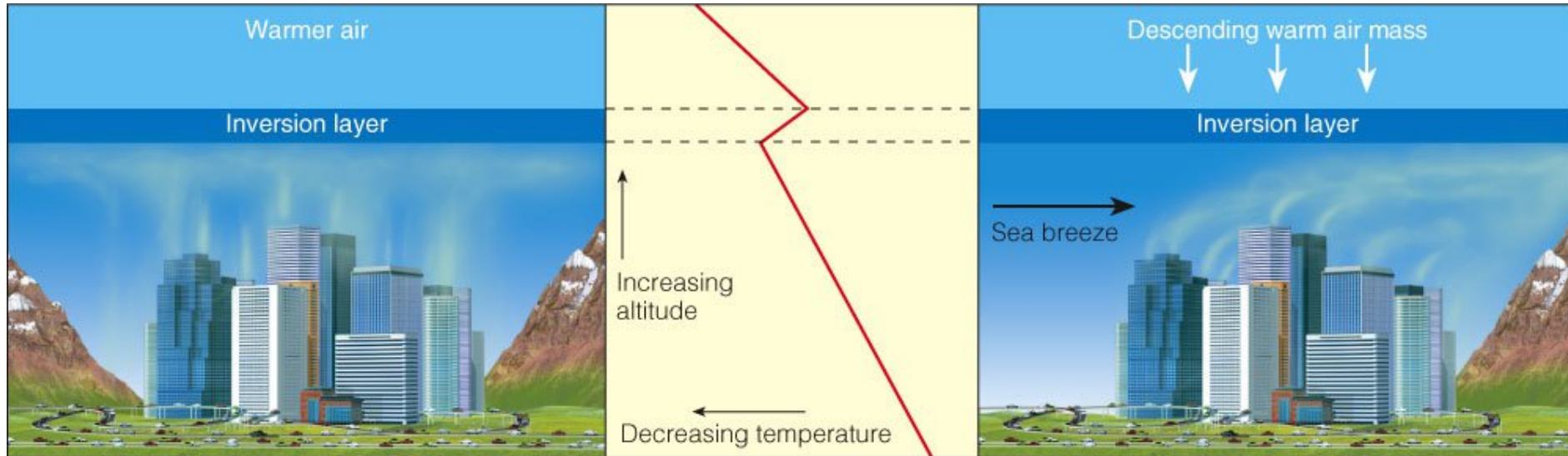
**Acid rain, or acid deposition,** is a broad term that includes any form of **precipitation** that contains **acidic** components, such as sulfuric **acid** or nitric **acid**, according to the Environmental Protection Agency (EPA).



Air pollution is one of several interacting stresses that can damage, weaken, or kill trees and pollute surface and groundwater.

Fig. 19-9, p. 451

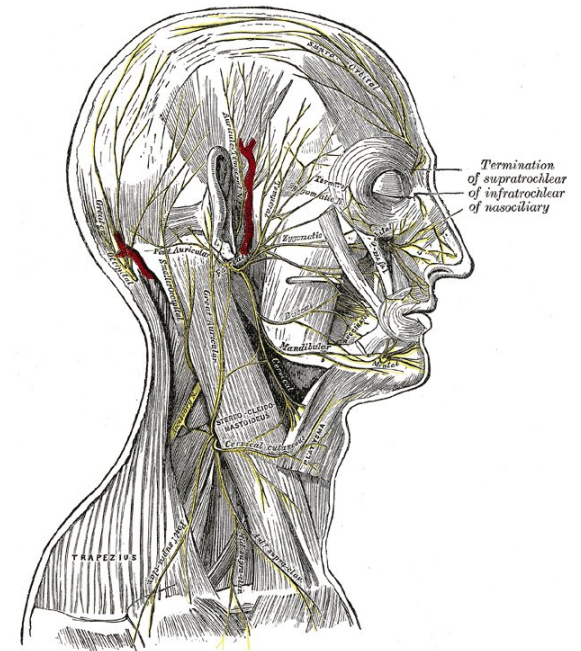
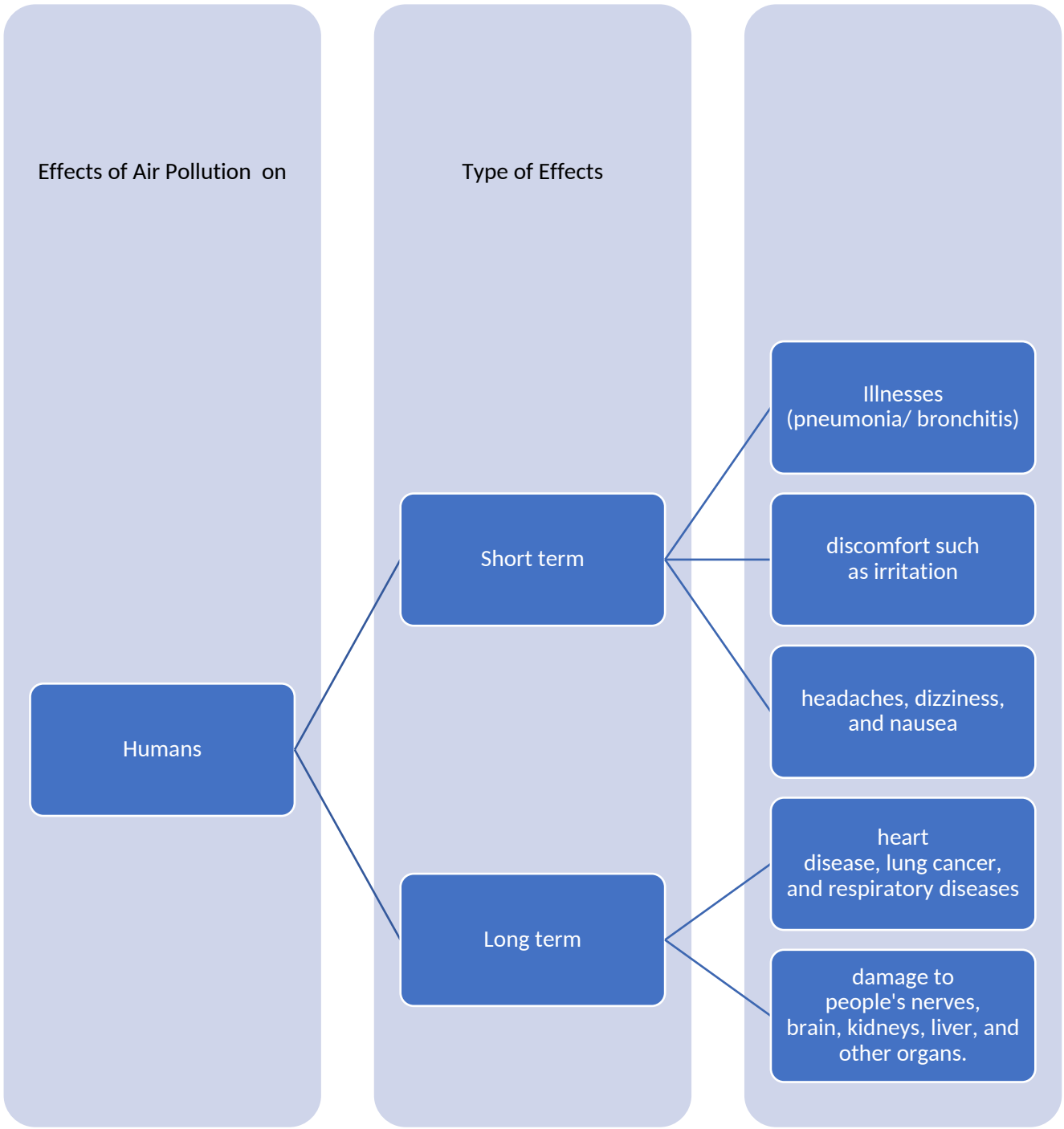
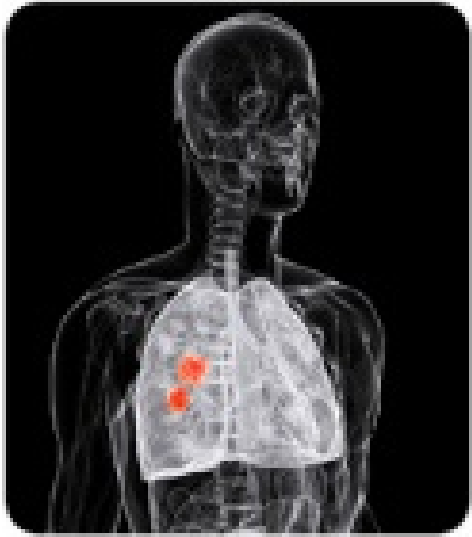
# Temperature Inversions – Effects of Air Pollution in Urban Areas



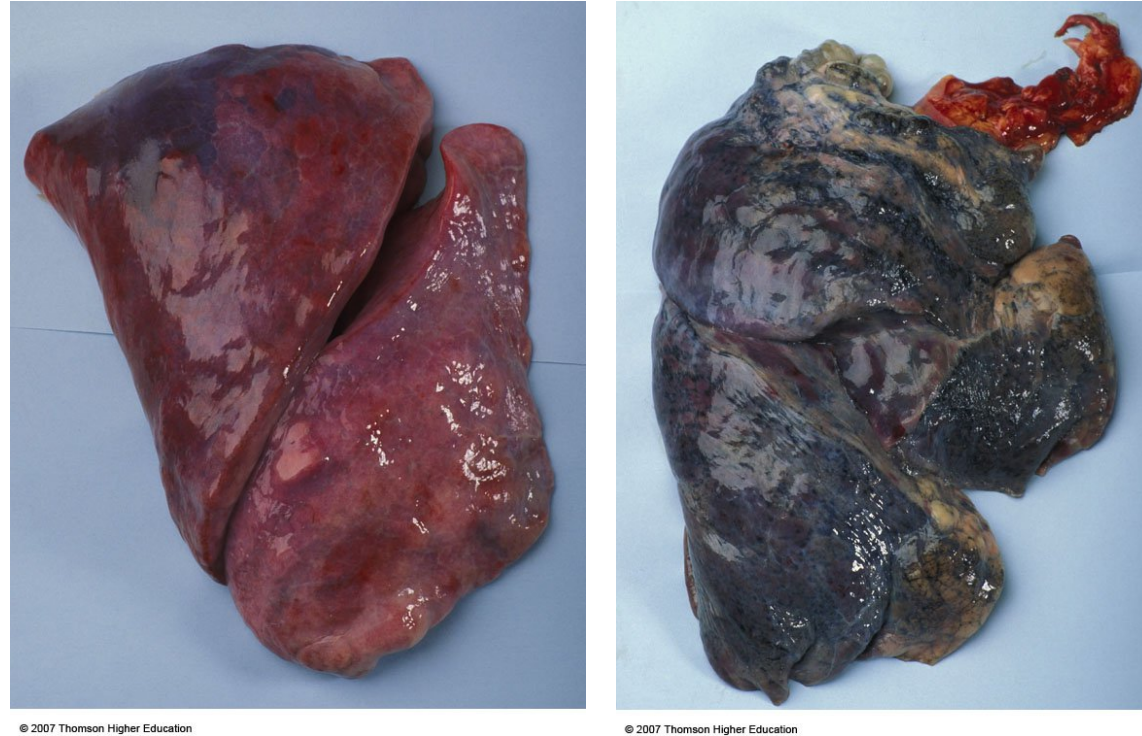
© 2007 Thomson Higher Education

- Cold, cloudy weather in a valley surrounded by mountains can trap air pollutants (left).
- Areas with sunny climate, light winds, mountains on three sides and an ocean on the other (right) are susceptible to inversions.

Figure 19-5



# HEALTH EFFECTS OF AIR POLLUTION



Normal human lungs (left) and the lungs of a person who died of emphysema (right).

## Solutions

### Air Pollution

Outdoor

Indoor

Improve energy efficiency to reduce fossil fuel use



Reduce poverty

Rely more on lower-polluting natural gas



Distribute cheap & efficient cookstoves or solar cookers to poor families in developing countries

Rely more on renewable energy (especially solar cells, wind, & solar-produced hydrogen)



Reduce or ban indoor smoking

Transfer technologies for latest energy efficiency, renewable energy, & pollution prevention to developing countries



Develop simple and cheap tests for indoor pollutants such as particulates, radon, and formaldehyde

# Can we play a role to **REDUCE** these problems?



Ride your bike



Tell your friends and family about pollution

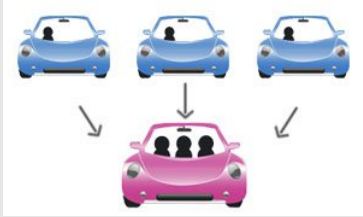


Make sure to conduct pollution checks at home



Take public transport

# Can we play a role to reduce these problems?



Carpooling



Switch off electric appliances when you leave room



Become an environmental scientist to do meaningful work

## Short Documentaries

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Bike Friendly City:- Copenhagen:

<https://www.youtube.com/watch?v=NAr5sB6aivk>

---

Take Public Bus- Vietnam:

<https://www.youtube.com/watch?v=lcmrEvkl8ps>

---

Carpooling in India:

<https://www.youtube.com/watch?v=vqYw1MRuHR0>

---

Traffic Control in Sweden:

<https://www.youtube.com/watch?v=fScV3mymDcA>



 alamy stock photo

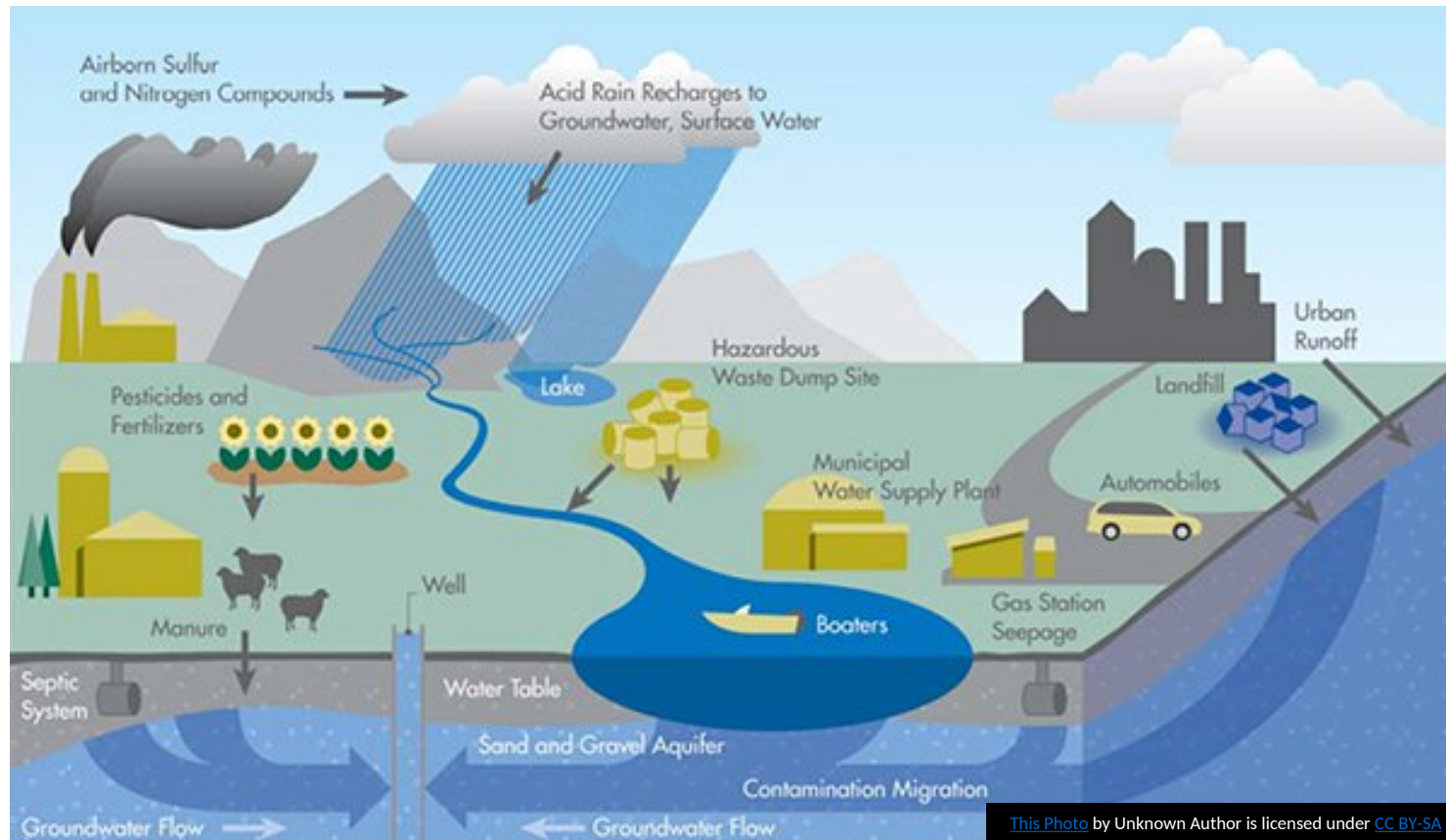
D4RMJ8  
www.alamy.com

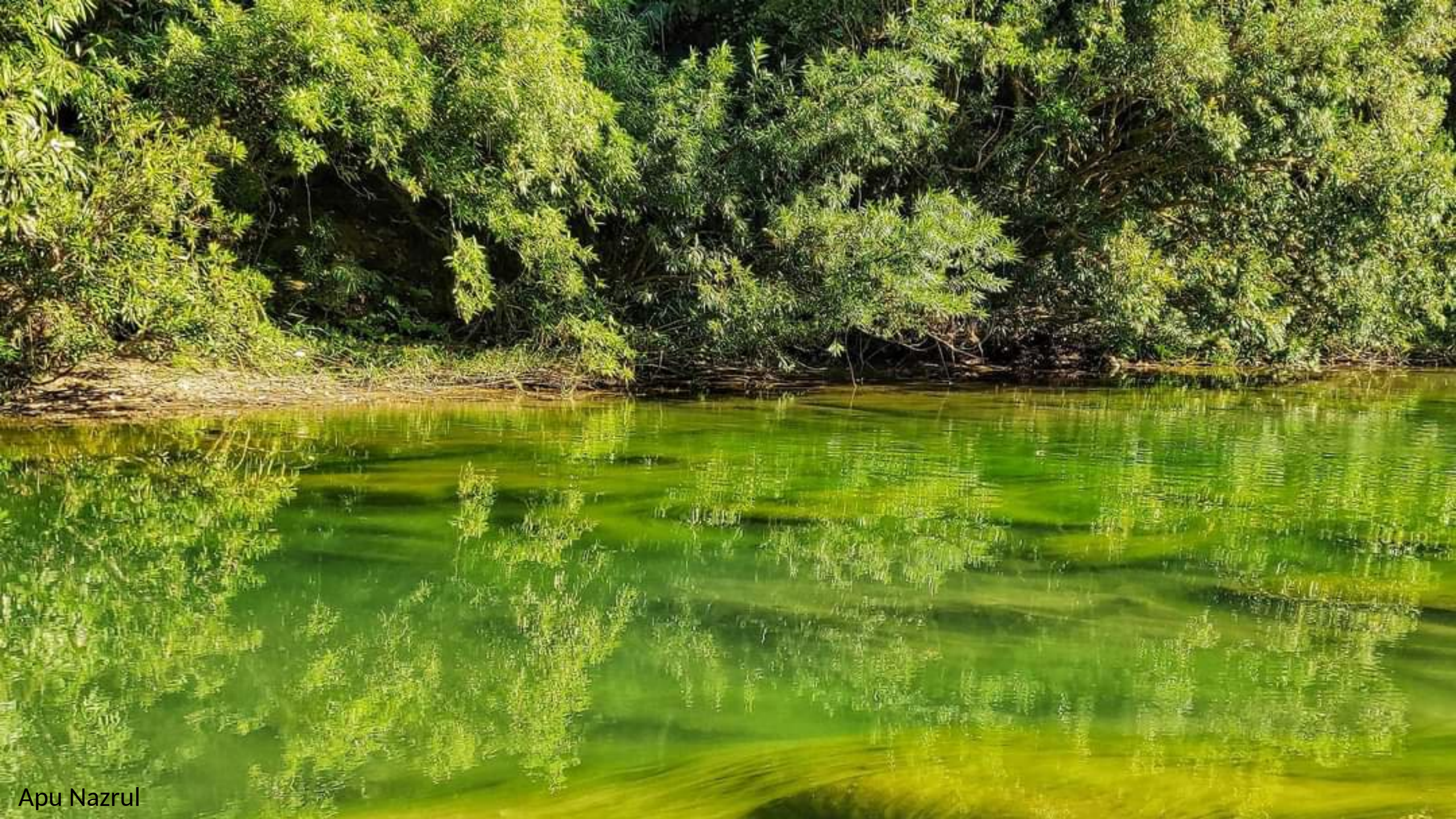
# Water Pollution:

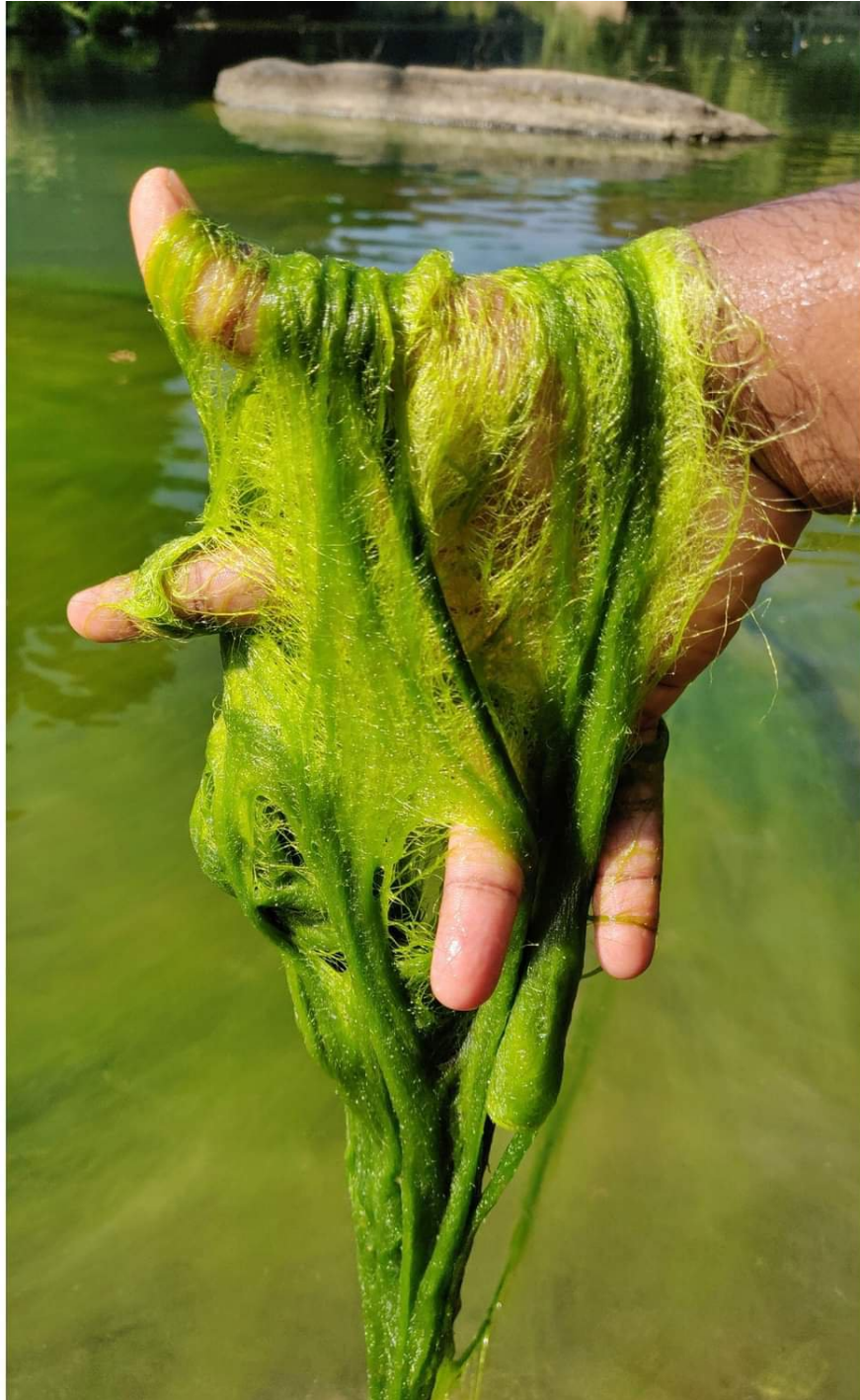
- ✓ Change in water quality that can harm organisms or make water unfit for human uses
- ✓ Contamination with chemicals
- ✓ Excessive heat

# Point

# Nonpoint









<https://images.app.goo.gl/7S4AT2eizKYTk6Bp9>



# Point & Nonpoint Source of Pollution



জেলা

সিরাজগঞ্জের বেলকুচি

## নলকূপের পানিতে বিষ, সুপেয় পানির সংকট

তামাই থামের নতুনপাড়ায় সুপেয় পানির সংকট বেশি। তাঁত কারখানার বর্জ্য ভূগর্ভস্থ পানির স্তরে মিশে দূষণ ঘটাচ্ছে বলে স্থানীয়দের ধারণা।



বৃহস্পতিবার, ০১ এপ্রিল ২০২১

 প্রথম আলো

# Major Water Pollutants and Their Sources

**Table 20-1** Major Water Pollutants and Their Sources

| Type/Effects                                                                         | Examples                                                            | Major Sources                                                                   |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Infectious agents (pathogens)<br><i>Cause diseases</i>                               | Bacteria, viruses, protozoa, parasites                              | Human and animal wastes                                                         |
| Oxygen-demanding wastes<br><i>Deplete dissolved oxygen needed by aquatic species</i> | Biodegradable animal wastes and plant debris                        | Sewage, animal feedlots, food-processing facilities, paper mills                |
| Plant nutrients<br><i>Cause excessive growth of algae and other species</i>          | Nitrates ( $\text{NO}_3^-$ ) and phosphates ( $\text{PO}_4^{3-}$ )  | Sewage, animal wastes, inorganic fertilizers                                    |
| Organic chemicals<br><i>Add toxins to aquatic systems</i>                            | Oil, gasoline, plastics, pesticides, fertilizers, cleaning solvents | Industry, farms, households, mining sites, runoff from streets and parking lots |
| Inorganic chemicals<br><i>Add toxins to aquatic systems</i>                          | Acids, bases, salts, metal compounds                                | Industry, households, mining sites, runoff from streets and parking lots        |
| Sediments<br><i>Disrupt photosynthesis, food webs, other processes</i>               | Soil, silt                                                          | Land erosion from farms and construction and mining sites                       |



Thank you